2025-09-12 - Analyze the parametric and non-parametric test data to solve the Analysis of Variance.

12 September 2025 15:36

Mod 5 : ANOVA (Non-Parametric)

-> The total variance in the joint-sample is being partitioned into 2 parts :

i) Between Samples Variance.

ii) Within Samples Variance.

=> Application :

-> B/w is used for different treatments.

-> Within is due to random unexplained disturbance.

=> $F_c = \dfrac{\text{B/w samples variance}}{\text{Within samples variance}}$.

=> H_0 : All population means are the same.

=> H_1 : All population means are not the same.

=> One-Way ANOVA.

i) Obtain mean of each sample : $\bar{x_1}, \bar{x_2}, \bar{x_3} \cdots \bar{x_k}$

ii) Workout the mean of sample means :

$$\bar{\bar{x}} = \frac{\bar{x_1} + \bar{x_2} + \bar{x_3} \cdots\cdots + \bar{x_k}}{\text{No. of samples } (k)}$$

iii) Take deviations of sample means from the mean of sample means & calculate the square of such deviations (Sum of Squared Between).

$$SSB = x_1\left(\bar{x_1} - \bar{\bar{x}}\right)^2 + x_2\left(\bar{x_2} - \bar{\bar{x}}\right)^2 + x_3\left(\bar{x_3} - \bar{\bar{x}}\right)^2 + \cdots + x_k\left(\bar{x_k} - \bar{\bar{x}}\right)^2$$

iv) Divide result of iii) by D-O-F b/w samples to obtain variance or mean square (MS) b/w.

$$MSB = \frac{SSB}{(k-1)}$$

v) Deviations within (SSW)

$$SSW = \sum\left(x_{1_i} - \bar{x_1}\right)^2 + \sum\left(x_{2_i} - \bar{x_2}\right)^2 + \cdots + \sum\left(x_{k_i} - \bar{x_k}\right)^2$$

vi) $MSW = \frac{SSW}{(n-k)}$

where $(n-k)$ represents D-o-F.

n: total number of items in all samples

i.e. $n_1 + n_2 + \cdots + n_k$

k = No. of samples.

vii) SS for Total Variance = $SSB + SSW$

DOF $(n-1) = (k-1) + (n-k)$

viii) F-Ratio $= \frac{MSB}{MSW}$

=> Set up an analysis of variance table for the following per acre production data for 3 varieties of wheat, each grown on 4 plot & state if the variety differences are significant.

	Per acre prod data		
Plot of land	Variety of wheat		
	A	B	C

	A	B	C
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

$\Rightarrow H_0$: All wheat varieties are same or effects of different varieties on wheat production are the same.

H_1 : All varieties are not the same.

i) $\overline{x_1} = \dfrac{6+7+3+8}{4} = 6$

$\overline{x_2} = \dfrac{5+5+3+7}{4} = 5$

$\overline{x_3} = \dfrac{5+4+3+4}{4} = 4$

ii) $\overline{\overline{x}} = \dfrac{\overline{x_1}+\overline{x_2}+\overline{x_3}}{k} = \dfrac{6+5+4}{3} = 5$

iii) $SSB = 4(6-5)^2 + 4(5-5)^2 + 4(4-5)^2$

$$= \underline{\underline{8}}$$

iv) $MSB = \dfrac{SSB}{(k-1)} = \dfrac{8}{2} = \underline{\underline{4}}$

v) $SSW = \Big[\{(6-6)^2 + (7-6)^2 + (3-6)^2 + (8-6)^2\}$
$+ \{(5-5)^2 + (5-5)^2 + (3-5)^2 + (7-5)^2\}$
$+ \{(5-4)^2 + (4-4)^2 + (3-4)^2 + (4-4)^2\}\Big]$

$= \underline{\underline{24}}$

vi) $MSW = \dfrac{SSW}{(n-k)} = \dfrac{24}{9} = \underline{\underline{2.67}}$

vii) $SST = SSB + SSW$
$= 8 + 24 = \underline{\underline{32}}$

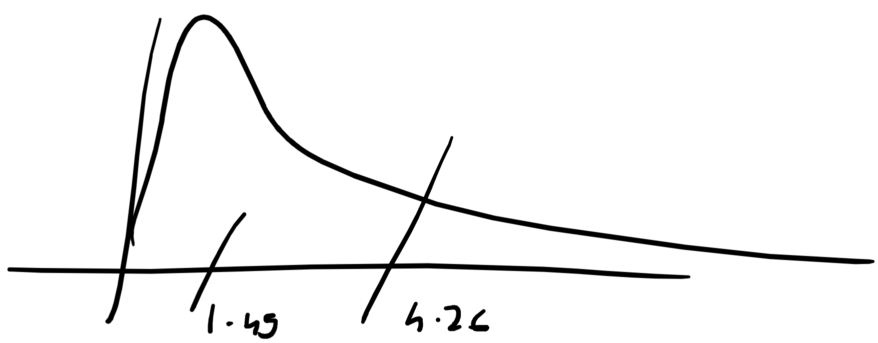
viii) $F\text{-Ratio} = \dfrac{MSB}{MSW} = \dfrac{4}{2.67} = \underline{\underline{1.49}}$

Anova Table:

Source	SS	DF	MS	F-Ratio	F_α
B/w Sample	8	$(3-1)=2$	4	1.49	$F(2, 9)$
Within Sample	24	$(12-3)=9$	2.67		$=4.26$

Within Sample	24	(12-3) = 9	2.67		=4.26
Total	32	(12-1) = 11			

	DF1	α = 0.05																	
DF2	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	Inf
1	161.45	199.5	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.1	251.14	252.2	253.25	254.31
2	18.513	19	19.164	19.247	19.296	19.33	19.353	19.371	19.385	19.396	19.413	19.429	19.446	19.454	19.462	19.471	19.479	19.487	19.496
3	10.128	9.5521	9.2766	9.1172	9.0135	8.9406	8.8867	8.8452	8.8123	8.7855	8.7446	8.7029	8.6602	8.6385	8.6166	8.5944	8.572	8.5494	8.5264
4	7.7086	6.9443	6.5914	6.3882	6.2561	6.1631	6.0942	6.041	5.9988	5.9644	5.9117	5.8578	5.8025	5.7744	5.7459	5.717	5.6877	5.6581	5.6281
5	6.6079	5.7861	5.4095	5.1922	5.0503	4.9503	4.8759	4.8183	4.7725	4.7351	4.6777	4.6188	4.5581	4.5272	4.4957	4.4638	4.4314	4.3985	4.365
6	5.9874	5.1433	4.7571	4.5337	4.3874	4.2839	4.2067	4.1468	4.099	4.06	3.9999	3.9381	3.8742	3.8415	3.8082	3.7743	3.7398	3.7047	3.6689
7	5.5914	4.7374	4.3468	4.1203	3.9715	3.866	3.787	3.7257	3.6767	3.6365	3.5747	3.5107	3.4445	3.4105	3.3758	3.3404	3.3043	3.2674	3.2298
8	5.3177	4.459	4.0662	3.8379	3.6875	3.5806	3.5005	3.4381	3.3881	3.3472	3.2839	3.2184	3.1503	3.1152	3.0794	3.0428	3.0053	2.9669	2.9276
9	5.1174	$\boxed{4.2565}$	3.8625	3.6331	3.4817	3.3738	3.2927	3.2296	3.1789	3.1373	3.0729	3.0061	2.9365	2.9005	2.8637	2.8259	2.7872	2.7475	2.7067
10	4.9646	4.1028	3.7083	3.478	3.3258	3.2172	3.1355	3.0717	3.0204	2.9782	2.913	2.845	2.774	2.7372	2.6996	2.6609	2.6211	2.5801	2.5379
11	4.8443	3.9823	3.5874	3.3567	3.2039	3.0946	3.0123	2.948	2.8962	2.8536	2.7876	2.7186	2.6464	2.609	2.5705	2.5309	2.4901	2.448	2.4045
12	4.7472	3.8853	3.4903	3.2592	3.1059	2.9961	2.9134	2.8486	2.7964	2.7534	2.6866	2.6169	2.5436	2.5055	2.4663	2.4259	2.3842	2.341	2.2962
13	4.6672	3.8056	3.4105	3.1791	3.0254	2.9153	2.8321	2.7669	2.7144	2.671	2.6037	2.5331	2.4589	2.4202	2.3803	2.3392	2.2966	2.2524	2.2064
14	4.6001	3.7389	3.3439	3.1122	2.9582	2.8477	2.7642	2.6987	2.6458	2.6022	2.5342	2.463	2.3879	2.3487	2.3082	2.2664	2.2229	2.1778	2.1307
15	4.5431	3.6823	3.2874	3.0556	2.9013	2.7905	2.7066	2.6408	2.5876	2.5437	2.4753	2.4034	2.3275	2.2878	2.2468	2.2043	2.1601	2.1141	2.0658
16	4.494	3.6337	3.2389	3.0069	2.8524	2.7413	2.6572	2.5911	2.5377	2.4935	2.4247	2.3522	2.2756	2.2354	2.1938	2.1507	2.1058	2.0589	2.0096
17	4.4513	3.5915	3.1968	2.9647	2.81	2.6987	2.6143	2.548	2.4943	2.4499	2.3807	2.3077	2.2304	2.1898	2.1477	2.104	2.0584	2.0107	1.9604
18	4.4139	3.5546	3.1599	2.9277	2.7729	2.6613	2.5767	2.5102	2.4563	2.4117	2.3421	2.2686	2.1906	2.1497	2.1071	2.0629	2.0166	1.9681	1.9168
19	4.3807	3.5219	3.1274	2.8951	2.7401	2.6283	2.5435	2.4768	2.4227	2.3779	2.308	2.2341	2.1555	2.1141	2.0712	2.0264	1.9795	1.9302	1.878
20	4.3512	3.4928	3.0984	2.8661	2.7109	2.599	2.514	2.4471	2.3928	2.3479	2.2776	2.2033	2.1242	2.0825	2.0391	1.9938	1.9464	1.8963	1.8432
21	4.3248	3.4668	3.0725	2.8401	2.6848	2.5727	2.4876	2.4205	2.366	2.321	2.2504	2.1757	2.096	2.054	2.0102	1.9645	1.9165	1.8657	1.8117
22	4.3009	3.4434	3.0491	2.8167	2.6613	2.5491	2.4638	2.3965	2.3419	2.2967	2.2258	2.1508	2.0707	2.0283	1.9842	1.938	1.8894	1.838	1.7831
23	4.2793	3.4221	3.028	2.7955	2.64	2.5277	2.4422	2.3748	2.3201	2.2747	2.2036	2.1282	2.0476	2.005	1.9605	1.9139	1.8648	1.8128	1.757
24	4.2597	3.4028	3.0088	2.7763	2.6207	2.5082	2.4226	2.3551	2.3002	2.2547	2.1834	2.1077	2.0267	1.9838	1.939	1.892	1.8424	1.7896	1.733
25	4.2417	3.3852	2.9912	2.7587	2.603	2.4904	2.4047	2.3371	2.2821	2.2365	2.1649	2.0889	2.0075	1.9643	1.9192	1.8718	1.8217	1.7684	1.711
26	4.2252	3.369	2.9752	2.7426	2.5868	2.4741	2.3883	2.3205	2.2655	2.2197	2.1479	2.0716	1.9898	1.9464	1.901	1.8533	1.8027	1.7488	1.6906
27	4.21	3.3541	2.9604	2.7278	2.5719	2.4591	2.3732	2.3053	2.2501	2.2043	2.1323	2.0558	1.9736	1.9299	1.8842	1.8361	1.7851	1.7306	1.6717
28	4.196	3.3404	2.9467	2.7141	2.5581	2.4453	2.3593	2.2913	2.236	2.19	2.1179	2.0411	1.9586	1.9147	1.8687	1.8203	1.7689	1.7138	1.6541
29	4.183	3.3277	2.934	2.7014	2.5454	2.4324	2.3463	2.2783	2.2229	2.1768	2.1045	2.0275	1.9446	1.9005	1.8543	1.8055	1.7537	1.6981	1.6376
30	4.1709	3.3158	2.9223	2.6896	2.5336	2.4205	2.3343	2.2662	2.2107	2.1646	2.0921	2.0148	1.9317	1.8874	1.8409	1.7918	1.7396	1.6835	1.6223
40	4.0847	3.2317	2.8387	2.606	2.4495	2.3359	2.249	2.1802	2.124	2.0772	2.0035	1.9245	1.8389	1.7929	1.7444	1.6928	1.6373	1.5766	1.5089
60	4.0012	3.1504	2.7581	2.5252	2.3683	2.2541	2.1665	2.097	2.0401	1.9926	1.9174	1.8364	1.748	1.7001	1.6491	1.5943	1.5343	1.4673	1.3893
120	3.9201	3.0718	2.6802	2.4472	2.2899	2.175	2.0868	2.0164	1.9588	1.9105	1.8337	1.7505	1.6587	1.6084	1.5543	1.4952	1.429	1.3519	1.2539
Inf	3.8415	2.9957	2.6049	2.3719	2.2141	2.0986	2.0096	1.9384	1.8799	1.8307	1.7522	1.6664	1.5705	1.5173	1.4591	1.394	1.318	1.2214	1



Verdict: Null Hypothesis is accepted.

=> Short-Cut Method for solving ANOVA :

i) Take the total & denote it by T.

ii) Workout the correction factor :

$$C_F = \frac{(T)^2}{n}$$

iii) Total SS = $\sum x_{ij}^2 - \frac{(T)^2}{n}$

iv) SSB = $\sum \frac{(T_j)^2}{n_j} - \frac{(T)^2}{n}$

v) SSW = $\left\{ \sum x_{ij}^2 - \frac{(T)^2}{n} \right\} - \left\{ \sum \frac{(T_j)^2}{n_j} - (T)^2/n \right\}$

$$\boxed{= \sum x_{ij}^2 - \sum \frac{(T_j)^2}{n_j}}$$

i) T = 60 ; n = 12

ii) $C_F = \frac{(T)^2}{n} = \frac{60 \times 60}{12} = 300$

iii) Total SS = $\sum x_{ij}^2 - C_F$

$$= \left[(6)^2 + (7)^2 + (3)^2 + (8)^2 + (5)^2 + (5)^2 + (3)^2 + (7)^2 + (5)^2 + (4)^2 + (3)^2 + (4)^2 \right] - CF$$

$$= 332 - 300 = \underline{\underline{32}}$$

iv) $SSB = \sum \frac{(T_j)^2}{n_j} - CF$

$$= \left(\frac{24 \times 24}{4}\right) + \left(\frac{20 \times 20}{4}\right) + \left(\frac{16 \times 16}{4}\right) - \left(\frac{60 \times 60}{12}\right)$$

$$= (144 + 100 + 64) - 300$$

$$= \underline{\underline{8}}$$

$SSW = \sum x_{ij}^2 - \sum \frac{(T_j)^2}{n_j}$

$$= 332 - 308 = \underline{\underline{24}}$$

⇒ **Two-Way ANOVA:** Data are classified on basis of 2 factors.

⇒ <u>One Observation per cell</u> :

→ Take the total of observations in all the samples & call it T.

→ Workout the correction factor : $\frac{(T)^2}{n}$

→ T SS

→ SS B_{cols}

→ SS B_{rows}

→ SSR (Residual or error variance)
= Total SS − SS B_{col} − SS B_{rows}

→ D-O-F
↳ D.F for total variance = $(c \cdot r - 1)$
↳ —— " —— B_{cols} = $(c-1)$
↳ —— " —— B_{rows} = $(r-1)$
↳ —— " —— residual variance = $(c-1) \cdot (r-1)$

Source	SS	DF	MS	F-Ratio
B/W columns	$\frac{\sum (T_j)^2}{n_j} - \frac{(T)^2}{n}$	$(c-1)$	$\frac{SSB_c}{(c-1)}$	$\frac{MSB_c}{MSR}$
B/W rows	$\frac{\sum (T_i)^2}{n_i} - \frac{(T)^2}{n}$	$(r-1)$	$\frac{SSB_r}{(r-1)}$	$\frac{MSB_r}{MSR}$
Residual / Error	$SST - (SSB_c + SSB_r)$	$(c-1)\cdot(r-1)$	$\frac{SSR}{(c-1)\cdot(r-1)}$	
Total	$\sum x_{ij}^2 - G =$	$(c \cdot r - 1)$		

i) H_{0c} : The varieties of first factor have same effect.

H_{1c} : The varieties of first factor are significantly different.

ii) H_{0R} : The varieties of second factor have the same effect.

H_{1R} : The varieties of second factor are significantly different.

$\Rightarrow F_{col} = \frac{MSB_{col}}{\quad} \quad \{(c-1), (c-1)\cdot(r-1)\}$

$$\Rightarrow F_{col} = \frac{MSB_{col}}{MSR} \quad \{(c-1), (c-1)\cdot(r-1)\}$$

$$\Rightarrow F_{row} = \frac{MSB_{rows}}{MSR} \quad \{(r-1), (c-1)\cdot(r-1)\}$$

⇒ Set up ANOVA table for 2-way design :

Varieties of fertilizers Variety of seeds

	A	B	C
W	6	5	5
X	7	5	4
Y	3	3	3
Z	8	7	4

⇒ i) $T = 60$; $n = 12$

ii) $C_F = \frac{(T)^2}{n} = \frac{60 \times 60}{12} = 300$

iii) $SST = \underline{\underline{32}}$

iv) $SSB_{col} = \left[\frac{24 \times 24}{4} + \frac{20 \times 20}{4} + \frac{16 \times 16}{4}\right] - \left[\frac{60 \times 60}{12}\right]$

$= 308 - 300 = \underline{\underline{8}}$

v) $SSB_{rows} = \left[\frac{16\times16}{3} + \frac{16\times16}{3} + \frac{9\times9}{3} + \frac{19\times19}{3}\right] - \left[\frac{60\times60}{12}\right]$

$= \underline{\underline{18}}$

vi) $SSR = SST - (SSB_{cols} + SSB_{rows})$

$= 32 - (8 + 18) = \underline{\underline{6}}$

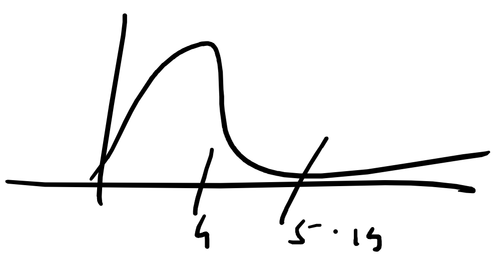
Source	SS	DF	MS	F-Ratio	F-α
B/w cols	8	$(c-1) = 2$	$\frac{8}{2} = 4$	$\frac{MSB_{col}}{MSR} = \frac{4}{1} = 4$	$F(2,6) = 5.14$
B/w rows	18	$(r-1) = 3$	$\frac{18}{3} = 6$	$\frac{MSB_{rows}}{MSR} = \frac{6}{1} = 6$	$F(3,6) = 4.76$
Residual	6	$(3-1)\cdot(4-1) = 6$	$\frac{6}{6} = 1$		
Error	32	$(3\times4-1) = 11$			

	DF1	α = 0.05																		
DF2	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	Inf
1	161.45	199.5	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.1	251.14	252.2	253.25	254.31
2	18.513	19	19.164	19.247	19.296	19.33	19.353	19.371	19.385	19.396	19.413	19.429	19.446	19.454	19.462	19.471	19.479	19.487	19.496
3	10.128	9.5521	9.2766	9.1172	9.0135	8.9406	8.8867	8.8452	8.8123	8.7855	8.7446	8.7029	8.6602	8.6385	8.6166	8.5944	8.572	8.5494	8.5264
4	7.7086	6.9443	6.5914	6.3882	6.2561	6.1631	6.0942	6.041	5.9988	5.9644	5.9117	5.8578	5.8025	5.7744	5.7459	5.717	5.6877	5.6581	5.6281
5	6.6079	5.7861	5.4095	5.1922	5.0503	4.9503	4.8759	4.8183	4.7725	4.7351	4.6777	4.6188	4.5581	4.5272	4.4957	4.4638	4.4314	4.3985	4.365
6	5.9874	5.1433	4.7571	4.5337	4.3874	4.2839	4.2067	4.1468	4.099	4.06	3.9999	3.9381	3.8742	3.8415	3.8082	3.7743	3.7398	3.7047	3.6689
7	5.5914	4.7374	4.3468	4.1203	3.9715	3.866	3.787	3.7257	3.6767	3.6365	3.5747	3.5107	3.4445	3.4105	3.3758	3.3404	3.3043	3.2674	3.2298
8	5.3177	4.459	4.0662	3.8379	3.6875	3.5806	3.5005	3.4381	3.3881	3.3472	3.2839	3.2184	3.1503	3.1152	3.0794	3.0428	3.0053	2.9669	2.9276
9	5.1174	4.2565	3.8625	3.6331	3.4817	3.3738	3.2927	3.2296	3.1789	3.1373	3.0729	3.0061	2.9365	2.9005	2.8637	2.8259	2.7872	2.7475	2.7067
10	4.9646	4.1028	3.7083	3.478	3.3258	3.2172	3.1355	3.0717	3.0204	2.9782	2.913	2.845	2.774	2.7372	2.6996	2.6609	2.6211	2.5801	2.5379
11	4.8443	3.9823	3.5874	3.3567	3.2039	3.0946	3.0123	2.948	2.8962	2.8536	2.7876	2.7186	2.6464	2.609	2.5705	2.5309	2.4901	2.448	2.4045
12	4.7472	3.8853	3.4903	3.2592	3.1059	2.9961	2.9134	2.8486	2.7964	2.7534	2.6866	2.6169	2.5436	2.5055	2.4663	2.4259	2.3842	2.341	2.2962
13	4.6672	3.8056	3.4105	3.1791	3.0254	2.9153	2.8321	2.7669	2.7144	2.671	2.6037	2.5331	2.4589	2.4202	2.3803	2.3392	2.2966	2.2524	2.2064
14	4.6001	3.7389	3.3439	3.1122	2.9582	2.8477	2.7642	2.6987	2.6458	2.6022	2.5342	2.463	2.3879	2.3487	2.3082	2.2664	2.2229	2.1778	2.1307
15	4.5431	3.6823	3.2874	3.0556	2.9013	2.7905	2.7066	2.6408	2.5876	2.5437	2.4753	2.4034	2.3275	2.2878	2.2468	2.2043	2.1601	2.1141	2.0658
16	4.494	3.6337	3.2389	3.0069	2.8524	2.7413	2.6572	2.5911	2.5377	2.4935	2.4247	2.3522	2.2756	2.2354	2.1938	2.1507	2.1058	2.0589	2.0096
17	4.4513	3.5915	3.1968	2.9647	2.81	2.6987	2.6143	2.548	2.4943	2.4499	2.3807	2.3077	2.2304	2.1898	2.1477	2.104	2.0584	2.0107	1.9604
18	4.4139	3.5546	3.1599	2.9277	2.7729	2.6613	2.5767	2.5102	2.4563	2.4117	2.3421	2.2686	2.1906	2.1497	2.1071	2.0629	2.0166	1.9681	1.9168
19	4.3807	3.5219	3.1274	2.8951	2.7401	2.6283	2.5435	2.4768	2.4227	2.3779	2.308	2.2341	2.1555	2.1141	2.0712	2.0264	1.9795	1.9302	1.878
20	4.3512	3.4928	3.0984	2.8661	2.7109	2.599	2.514	2.4471	2.3928	2.3479	2.2776	2.2033	2.1242	2.0825	2.0391	1.9938	1.9464	1.8963	1.8432
21	4.3248	3.4668	3.0725	2.8401	2.6848	2.5727	2.4876	2.4205	2.366	2.321	2.2504	2.1757	2.096	2.054	2.0102	1.9645	1.9165	1.8657	1.8117

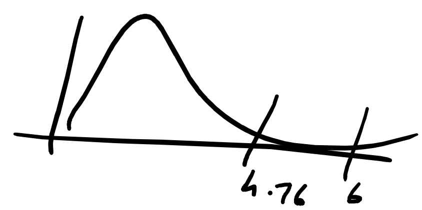
18	4.4139	3.5546	3.1599	2.9277	2.7729	2.6613	2.5767	2.5102	2.4563	2.4117	2.3421	2.2686	2.1906	2.1497	2.1071	2.0629	2.0166	1.9681	1.9168
19	4.3807	3.5219	3.1274	2.8951	2.7401	2.6283	2.5435	2.4768	2.4227	2.3779	2.308	2.2341	2.1555	2.1141	2.0712	2.0264	1.9795	1.9302	1.878
20	4.3512	3.4928	3.0984	2.8661	2.7109	2.599	2.514	2.4471	2.3928	2.3479	2.2776	2.2033	2.1242	2.0825	2.0391	1.9938	1.9464	1.8963	1.8432
21	4.3248	3.4668	3.0725	2.8401	2.6848	2.5727	2.4876	2.4205	2.366	2.321	2.2504	2.1757	2.096	2.054	2.0102	1.9645	1.9165	1.8657	1.8117
22	4.3009	3.4434	3.0491	2.8167	2.6613	2.5491	2.4638	2.3965	2.3419	2.2967	2.2258	2.1508	2.0707	2.0283	1.9842	1.938	1.8894	1.838	1.7831
23	4.2793	3.4221	3.028	2.7955	2.64	2.5277	2.4422	2.3748	2.3201	2.2747	2.2036	2.1282	2.0476	2.005	1.9605	1.9139	1.8648	1.8128	1.757
24	4.2597	3.4028	3.0088	2.7763	2.6207	2.5082	2.4226	2.3551	2.3002	2.2547	2.1834	2.1077	2.0267	1.9838	1.939	1.892	1.8424	1.7896	1.733
25	4.2417	3.3852	2.9912	2.7587	2.603	2.4904	2.4047	2.3371	2.2821	2.2365	2.1649	2.0889	2.0075	1.9643	1.9192	1.8718	1.8217	1.7684	1.711
26	4.2252	3.369	2.9752	2.7426	2.5868	2.4741	2.3883	2.3205	2.2655	2.2197	2.1479	2.0716	1.9898	1.9464	1.901	1.8533	1.8027	1.7488	1.6906
27	4.21	3.3541	2.9604	2.7278	2.5719	2.4591	2.3732	2.3053	2.2501	2.2043	2.1323	2.0558	1.9736	1.9299	1.8842	1.8361	1.7851	1.7306	1.6717
28	4.196	3.3404	2.9467	2.7141	2.5581	2.4453	2.3593	2.2913	2.236	2.19	2.1179	2.0411	1.9586	1.9147	1.8687	1.8203	1.7689	1.7138	1.6541
29	4.183	3.3277	2.934	2.7014	2.5454	2.4324	2.3463	2.2783	2.2229	2.1768	2.1045	2.0275	1.9446	1.9005	1.8543	1.8055	1.7537	1.6981	1.6376
30	4.1709	3.3158	2.9223	2.6896	2.5336	2.4205	2.3343	2.2662	2.2107	2.1646	2.0921	2.0148	1.9317	1.8874	1.8409	1.7918	1.7396	1.6835	1.6223
40	4.0847	3.2317	2.8387	2.606	2.4495	2.3359	2.249	2.1802	2.124	2.0772	2.0035	1.9245	1.8389	1.7929	1.7444	1.6928	1.6373	1.5766	1.5089
60	4.0012	3.1504	2.7581	2.5252	2.3683	2.2541	2.1665	2.097	2.0401	1.9926	1.9174	1.8364	1.748	1.7001	1.6491	1.5943	1.5343	1.4673	1.3893
120	3.9201	3.0718	2.6802	2.4472	2.2899	2.175	2.0868	2.0164	1.9588	1.9105	1.8337	1.7505	1.6587	1.6084	1.5543	1.4952	1.429	1.3519	1.2539
Inf	3.8415	2.9957	2.6049	2.3719	2.2141	2.0986	2.0096	1.9384	1.8799	1.8307	1.7522	1.6664	1.5705	1.5173	1.4591	1.394	1.318	1.2214	1

Verdict:

Seeds



Fertilizer



∴ Accept null hypothesis for seeds & reject the null hypothesis for fertilizers.